

Forward Deployed Engineer

Build Case

THE SETTING

You are building for a vascular surgery practice in Florida. More than half of its patients are covered by Blue Cross Blue Shield of Florida (BCBS FL). The practice loses staff time getting procedures approved before they are performed, and loses revenue when those procedures are denied.

BACKGROUND

Before an insurer will pay for many procedures, the provider has to show in advance that the patient meets the insurer's criteria for it. This is prior authorization: staff must prove medical necessity by submitting a request to the insurer with supporting evidence drawn from the patient's chart. The criteria live in published medical necessity guidelines: long, dense documents that vary by insurer and procedure and change over time.

In practice, most staff do not read these guidelines directly. They rely on tribal knowledge: they broadly know which procedures insurers deny and why, so they read the patient's chart, infer what is missing, and try to fill the gaps before submitting. That knowledge lives in people's heads, it is uneven across staff, and when something is missed, the request is denied.

YOUR MATERIALS

Two things to root yourself on. Both are examples, not boundaries.

An example guideline • BCBS of Florida

A good guideline to root your product on. We will test the product against other guidelines, so it must let us upload a new guideline as a PDF and work against it.

[Treatments for Varicose Veins / Venous Insufficiency \(02-33000-31\)](#)

An example patient record • attached chart

This chart shows the format and depth of a patient chart. You can assume we can always get a PDF version of the patient chart. Feel free to create your own synthetic patients with different medical scenarios and orders.

[robert_mitchell_chart_final.pdf](#)

BUILD

Build a simple MVP that works: something that saves clinic staff time and reduces denials.

This must be a working product, not a demo. It has to run on real inputs, including patients it has not seen before. We will run it on ours. Keep it simple. A thin thing that runs end to end beats a broad thing that does not. The scope is up to you to define.

OUT OF SCOPE

Do not spend your time on compliance, security, or infrastructure. Specifically, ignore HIPAA and PHI handling.

HOW WE WILL EVALUATE IT

Send us your code one day before the call. Beforehand we will run it ourselves and feed it guidelines and patient records it has not seen, to find where it holds and where it breaks. The call is not a demo. It is a conversation about what we found: what you assumed, where the logic lives, why it broke where it did, and how you would extend it. Come ready to defend your tradeoffs.

CONSTRAINTS

- **It must run on our machine from a one-line command.** Test this on a clean checkout.
- **Any stack.** Ours is Python, FastAPI, React, TypeScript, Postgres.
- **Use AI coding tools freely.** We do.

DELIVERABLES

- Code (repo or zip) with a one-line run command, sent one day before the call.
- A short design note: where you put the judgment (what is data, what is code), and what breaks first under a guideline change or a messy chart.